

Lecture Notes on NAND Flash Memory Technology

Chapter 2: NAND Flash Architecture

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1 2.1 Cell Types and Density Evolution

- **Single-Level Cell (SLC)**

Definition 1.1 (SLC Characteristics). • 1 bit per cell (2 states)

- Highest reliability: $> 10^5$ P/E cycles
- Fastest programming: ~ 200 s/page
- Lowest density and highest cost/bit

- **Multi-Level Cell Types**

Bit capacity and state distribution:

$$\text{States} = 2^n, \quad \text{Margin} = \frac{V_{\text{window}}}{2^n - 1} \quad (1)$$

- **MLC**: 2 bits/cell, 4 states, $\sim 10^3$ cycles
- **TLC**: 3 bits/cell, 8 states, $\sim 10^2 - 10^3$ cycles
- **QLC**: 4 bits/cell, 16 states, $\sim 10^2$ cycles

2 2.2 NAND String Architecture

- **Basic String Structure**

NAND string components:

- N memory cells in series (typically 32, 64, or 128)
- String Select Line (SSL) transistor
- Ground Select Line (GSL) transistor

String current equation:

$$I_{\text{string}} = \frac{\mu C_{ox} W}{N \cdot L_{eff}} \left(\prod_{i=1}^N \frac{V_{gs,i} - V_{thi}}{V_{ds,total}/N} \right) V_{ds,total} \quad (2)$$

- **Select Gate Transistors**

Function of select gates:

$$\begin{cases} \text{SSL} : & \text{Connect string to bit line} \\ \text{GSL} : & \text{Connect string to source line} \end{cases} \quad (3)$$

Overdrive voltage requirement:

$$V_{SSL} > V_{thSSL} + V_{BL} + \Delta V_{margin} \quad (4)$$

3 2.3 Array Organization

- **Hierarchical Structure**

Theorem 3.1 (Array Hierarchy).

$$\text{Die} \rightarrow \text{Plane} \rightarrow \text{Block} \rightarrow \text{Page} \rightarrow \text{Byte} \quad (5)$$

$$\text{Capacity} = N_{plane} \times N_{block} \times N_{page} \times \text{Page size} \quad (6)$$

Typical parameters:

- Page size: 4KB, 8KB, 16KB
- Pages per block: 128, 256, 512
- Blocks per plane: 2048, 4096
- Planes per die: 2, 4, 8

- **Page Architecture**

Page composition:

$$\text{Page size} = \text{Main area} + \text{Spare area} \quad (7)$$

Example: 4KB + 224B spare for ECC and metadata

4 2.4 Bit Line and Word Line Organization

- **Bit Line Architecture**

Even-odd bit line arrangement:

$$\begin{cases} BL_{even} : & BL_0, BL_2, BL_4, \dots \\ BL_{odd} : & BL_1, BL_3, BL_5, \dots \end{cases} \quad (8)$$

Shielding effect:

$$C_{coupling} = \frac{\epsilon_0 \epsilon_r A}{d_{BL-BL}} \quad (9)$$

- **Word Line Arrangement**

RC delay model:

$$\tau_{WL} = R_{WL} \cdot C_{WL} = \frac{\rho \cdot L_{WL}^2}{t_{WL} \cdot w_{WL}} \cdot \epsilon_{ox} \frac{L_{WL} \cdot w_{WL}}{t_{ox}} \quad (10)$$

Word line driver requirements:

$$I_{driver} > C_{WL} \cdot \frac{dV}{dt} = C_{WL} \cdot \frac{V_{pgm}}{t_{rise}} \quad (11)$$

5 2.5 Peripheral Circuits

- **Row Decoder**

Address decoding:

$$N_{WL} = 2^{n_{row}}, \quad n_{row} = \log_2(N_{WL}) \quad (12)$$

Decoder power consumption:

$$P_{decoder} = \alpha \cdot f \cdot C_{load} \cdot V_{DD}^2 \quad (13)$$

- **Column Decoder and Page Buffer**

Page buffer functions:

- Sensing and latching read data
- Holding program data
- Verify operations
- Column redundancy

Sense amplifier equation:

$$\Delta V_{sense} = \frac{I_{cell} \cdot t_{sense}}{C_{BL}} \quad (14)$$

- **Charge Pumps**

Dickson charge pump output:

$$V_{out} = V_{in} + N_{stage} \cdot (V_{clock} - V_{diode}) - I_{load} \cdot \frac{N_{stage}}{f \cdot C} \quad (15)$$

Efficiency:

$$\eta = \frac{P_{out}}{P_{in}} = \frac{I_{out} \cdot V_{out}}{I_{in} \cdot V_{in}} \quad (16)$$

6 2.6 Layout Considerations

• Cell Area Efficiency

Effective cell size:

$$A_{cell} = 4F^2 \text{ (2D NAND)} \quad (17)$$

where F = minimum feature size

Array efficiency:

$$\eta_{array} = \frac{N_{cell} \cdot A_{cell}}{A_{total}} = \frac{A_{array}}{A_{array} + A_{peripheral}} \quad (18)$$

• Parasitic Capacitances

Total bit line capacitance:

$$C_{BL} = C_{junction} \cdot N_{cell} + C_{metal} \cdot L_{BL} + C_{coupling} \quad (19)$$

Impact on sensing:

$$t_{sense} = \frac{C_{BL} \cdot \Delta V_{min}}{I_{cell,min}} \quad (20)$$

7 2.7 Advanced Architectures

• All-Bit-Line (ABL) Architecture

Advantages:

- Simultaneous sensing of all bit lines
- Higher throughput: $2\times$ for read
- Better noise immunity

Page buffer requirements:

$$N_{PB} = N_{BL} = \frac{\text{Page size} \times 8}{\text{Bits per cell}} \quad (21)$$

• Multi-Plane Operations

Parallel operation efficiency:

$$\text{Throughput}_{multi} = N_{plane} \cdot \text{Throughput}_{single} \cdot \eta_{parallel} \quad (22)$$

where $\eta_{parallel} \approx 0.8 - 0.9$ due to overhead

8 Examples

Example 8.1 (Array Capacity Calculation). Calculate total capacity for: - 4 planes, 4096 blocks/plane
- 256 pages/block, 16KB page size - TLC (3 bits/cell)

Solution:

$$\text{Capacity} = 4 \times 4096 \times 256 \times 16\text{KB} \times 3/8 \quad (23)$$

$$= 4 \times 4096 \times 256 \times 16 \times 3/8 \text{ KB} \quad (24)$$

$$= 25,165,824 \text{ KB} \quad (25)$$

$$= 24 \text{ GB} \quad (26)$$

Example 8.2 (RC Delay Calculation). Calculate WL delay for:

Length: 5mm, Width: 50nm, Thickness: 100nm

Sheet resistance: 10 /

Capacitance: 1 fF/m

Solution:

$$R_{WL} = R_s \cdot \frac{L}{W} = 10 \times \frac{5000}{0.05} = 10^6 \quad (27)$$

$$C_{WL} = 1 \text{ fF/m} \times 5000 \text{ m} = 5 \text{ pF} \quad (28)$$

$$\tau = RC = 10^6 \times 5 \times 10^{-12} = 5 \text{ s} \quad (29)$$

9 Homework Problems

Assignment 2.1: NAND Architecture

1. String Design:

- Design 128-cell NAND string
- Calculate total resistance and capacitance
- Determine optimal V_{pass} for $5V V_{\text{th}}$ distribution
- Analyze current degradation

2. Array Efficiency:

- 512Gb die with 16KB pages
- Calculate number of blocks needed
- Design plane architecture for optimal parallelism
- Estimate peripheral area overhead (15%)

3. Bit Line Coupling:

- BL pitch: 40nm, height: 300nm
- Calculate coupling capacitance
- Design shielding scheme

- Analyze noise margin impact
4. **Page Buffer Design:**
- Design latch circuit for MLC operation
 - Support cache programming
 - Calculate area and power
 - Optimize for 200MB/s throughput
5. **Charge Pump Optimization:**
- Generate 20V from 3.3V supply
 - Load current: 10mA
 - Design Dickson pump stages
 - Calculate efficiency and area
6. **Multi-Die Package:**
- 8-die stack with shared I/O
 - Design chip enable and ready/busy signals
 - Calculate interface bandwidth requirements
 - Optimize for queue depth = 32
7. **Decoder Complexity:**
- 8192 word lines per block
 - Design hierarchical decoder
 - Calculate transistor count
 - Estimate propagation delay
8. **Advanced Architecture:**
- Compare ABL vs shared BL schemes
 - Analyze area, speed, power trade-offs
 - Design for 1GB/s read throughput
 - Include ECC overhead calculations

Next Chapter Preview: Chapter 3 will cover "Programming and Erasing Operations" including ISPP, erase algorithms, and disturb mechanisms.